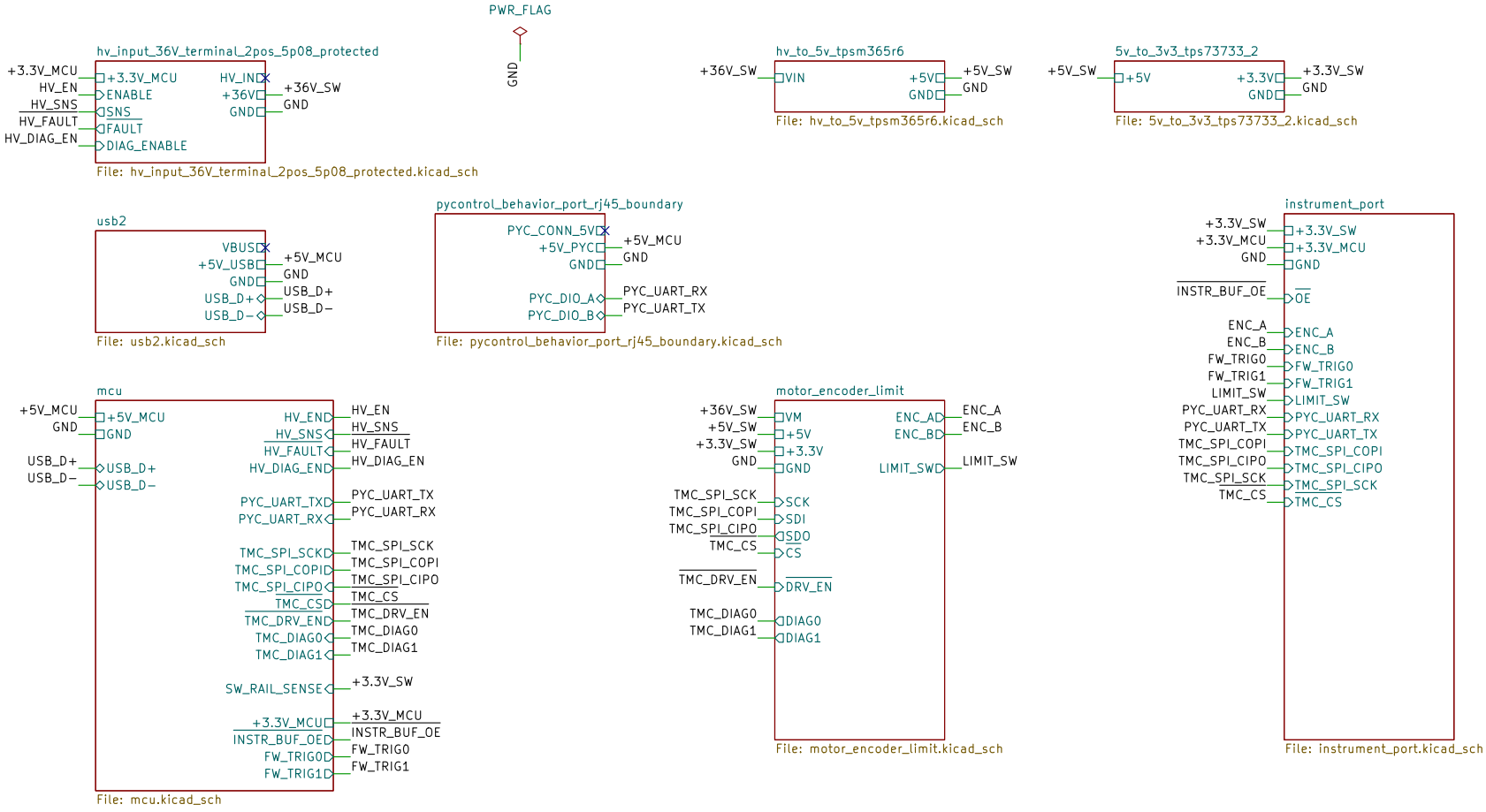


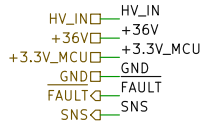
Janelia Research Campus

Sheet: /

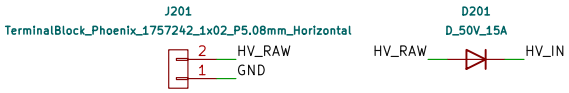
File: mouse−joystick−pcb.kicad_sch

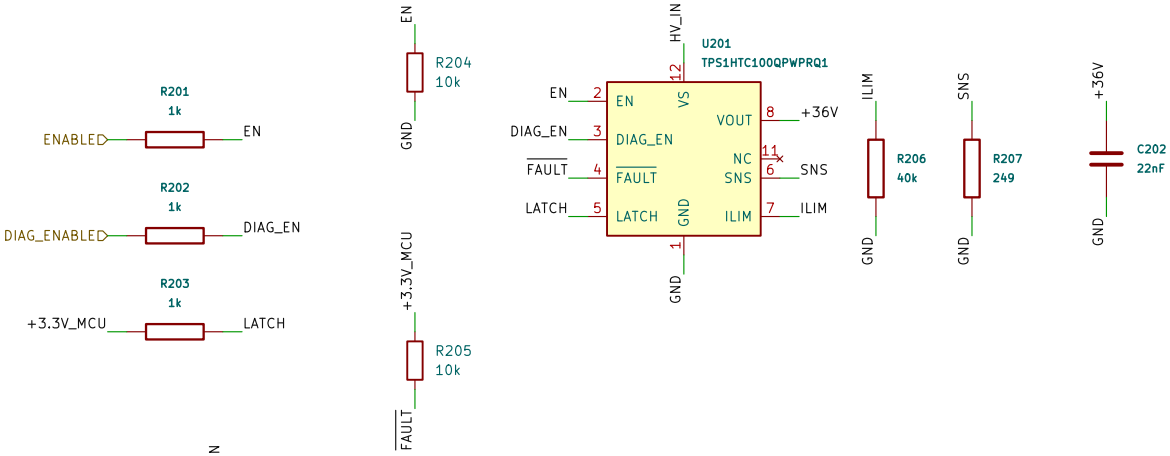
Title: mouse−joystick−pcb

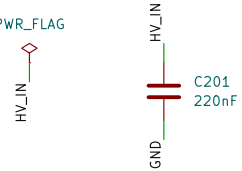
Size: A4	Date: 2026−08−19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 1/19



R_ILIM	I_CL min	typ	max	source
49.9k / 50k	0.73 A	0.92 A	1.11 A	datasheet
40k	0.9 A	1.2 A	1.4 A	computed
25k	1.5 A	1.96 A	2.3 A	datasheet
20k	1.9 A	2.5 A	2.9 A	computed
16.7k	2.3 A	3.0 A	3.5 A	datasheet
10k	4.0 A	5.3 A	6.3 A	datasheet
open or GND	6.1 A	7.3 A	8.6 A	datasheet — do not use







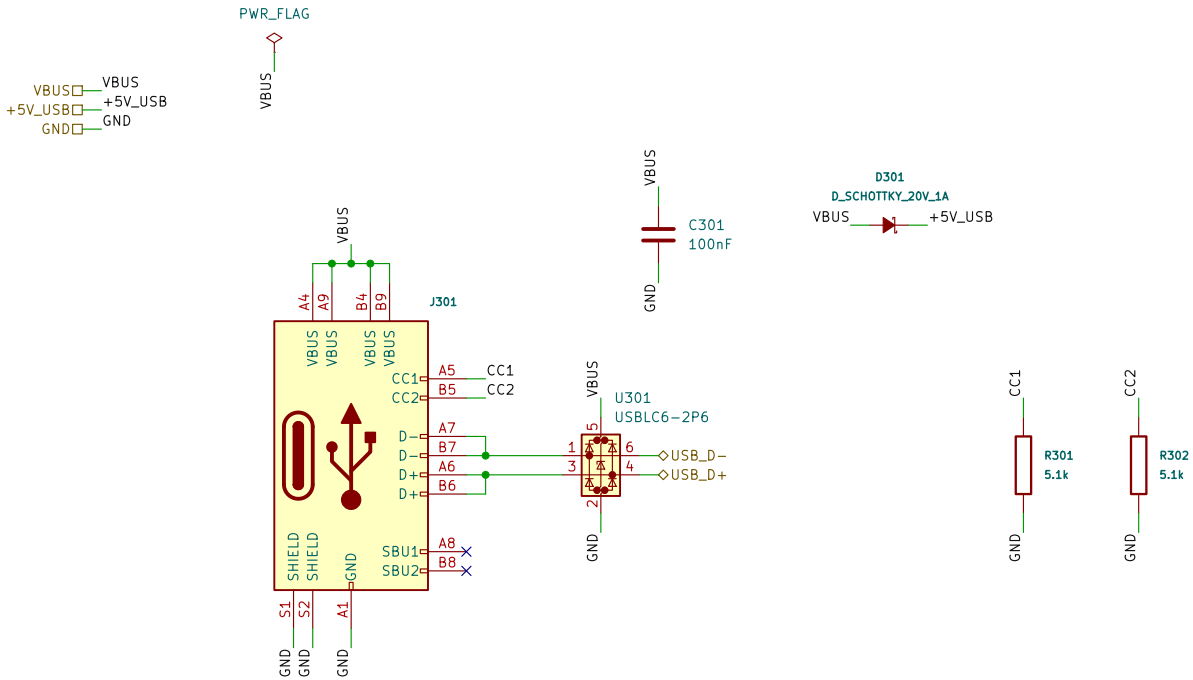
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

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Sheet: /hv_input_36V_terminal_2pos_5p08_protected/
File: hv_input_36V_terminal_2pos_5p08_protected.kicad_sch

Title: mouse-joystick-pcb

Size: A4	Date: 2026-08-19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 2/19



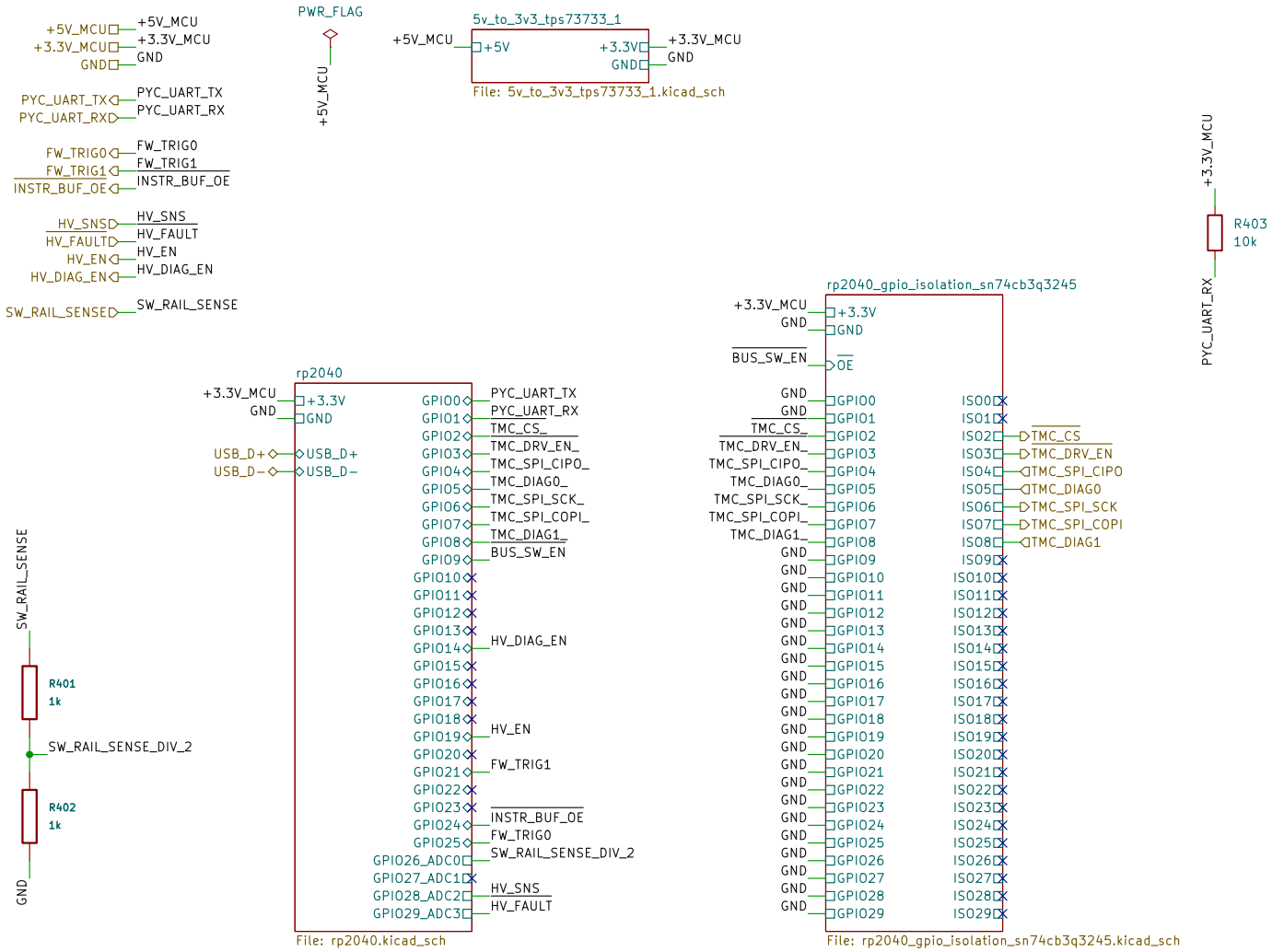
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

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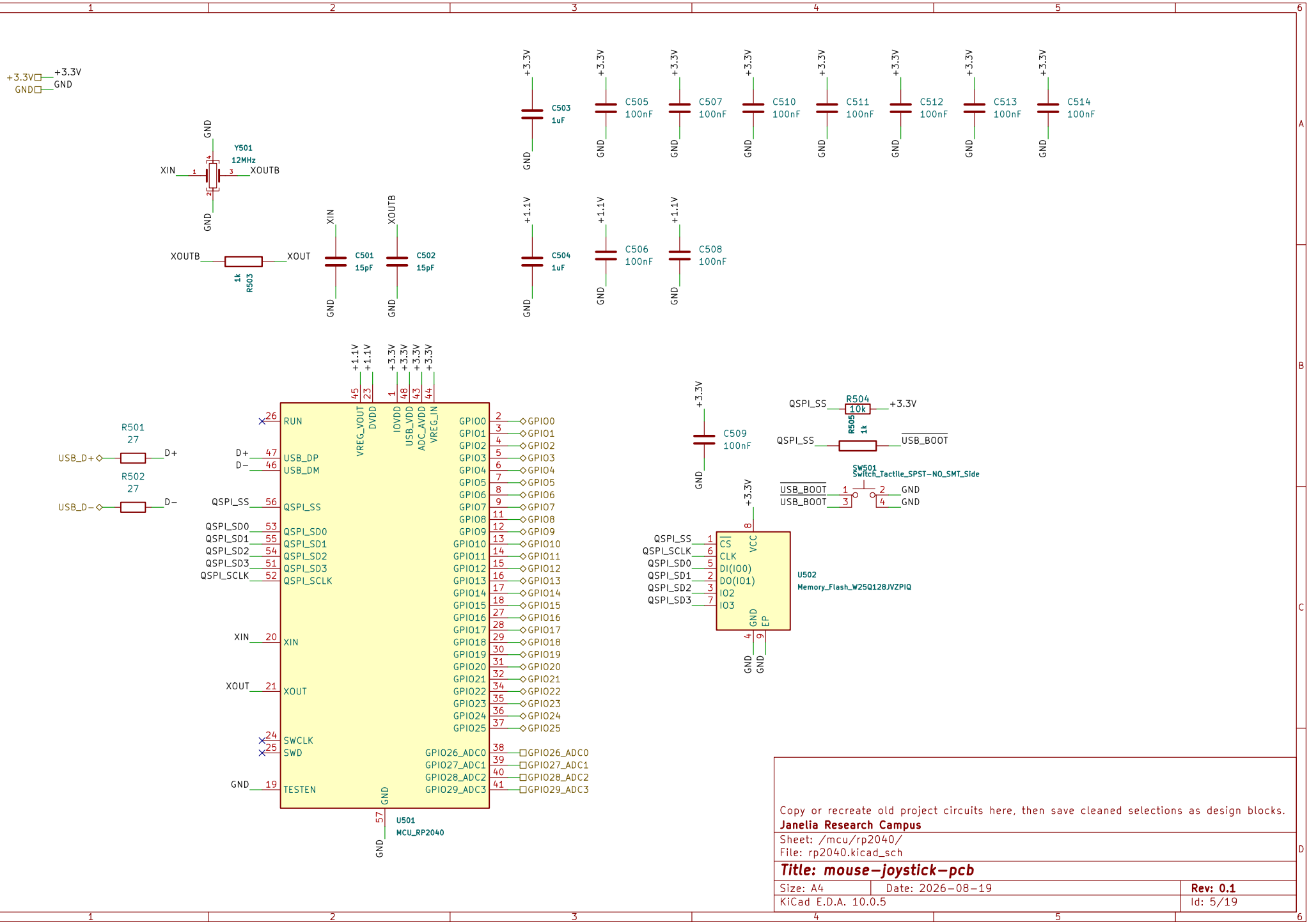
Sheet: /usb2/
File: usb2.kicad_sch

Title: mouse−joystick−pcb

Size: A4	Date: 2026−08−19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 3/19



Janelia Research Campus		
Sheet: /mcu/		
File: mcu.kicad_sch		
Title: mouse−joystick−pcb		
Size: A4	Date: 2026−08−19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 4/19



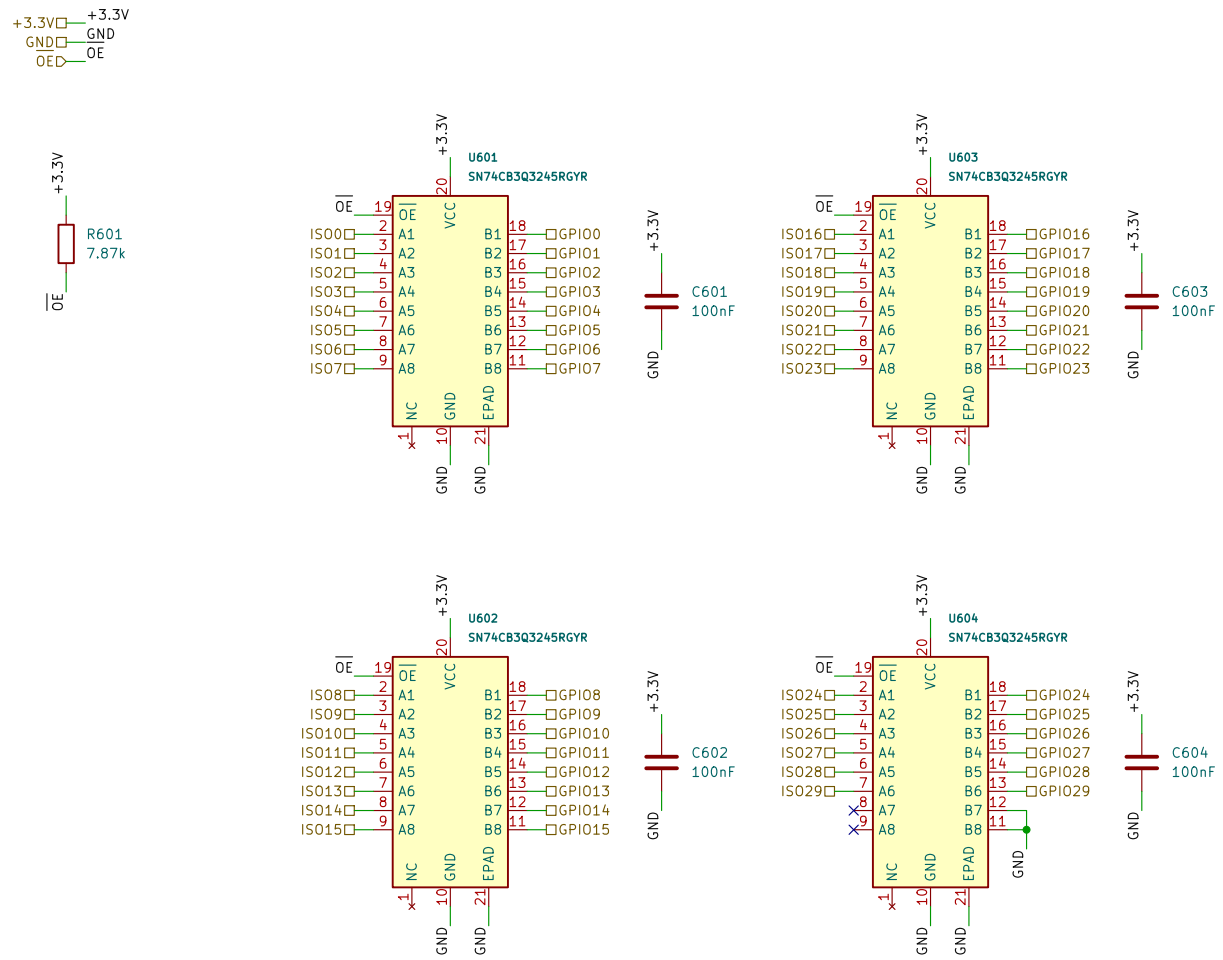
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

Janelia Research Campus

Sheet: /mcu/rp2040/
File: rp2040.kicad_sch

Title: mouse-joystick-pcb

Size: A4	Date: 2026-08-19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 5/19



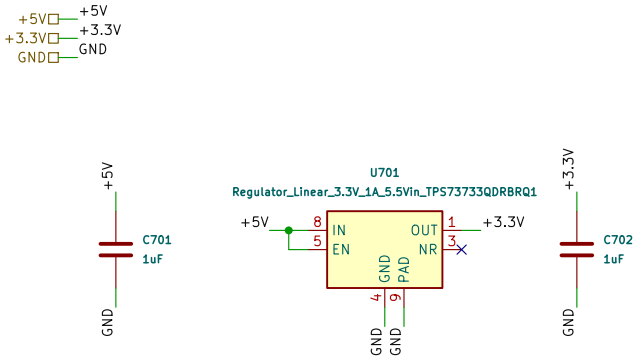
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

Janelia Research Campus

Sheet: /mcu/rp2040_gpio_isolation_sn74cb3q3245/
File: rp2040_gpio_isolation_sn74cb3q3245.kicad_sch

Title: mouse-joystick-pcb

Size: A4	Date: 2026-08-19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 6/19



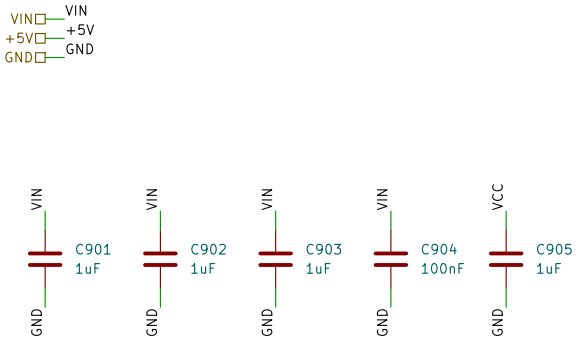
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

Janelia Research Campus

Sheet: /mcu/5v_to_3v3_tps73733_1/
File: 5v_to_3v3_tps73733_1.kicad_sch

Title: mouse-joystick-pcb

Size: A4	Date: 2026-08-19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 7/19



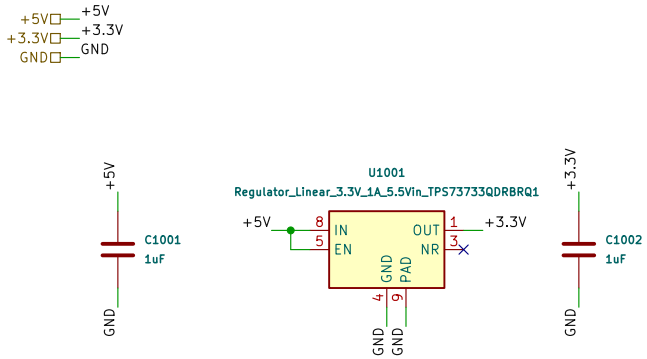
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

Janelia Research Campus

Sheet: /hv_to_5v_tpsm365r6/
File: hv_to_5v_tpsm365r6.kicad_sch

Title: mouse-joystick-pcb

Size: A4	Date: 2026-08-19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 9/19



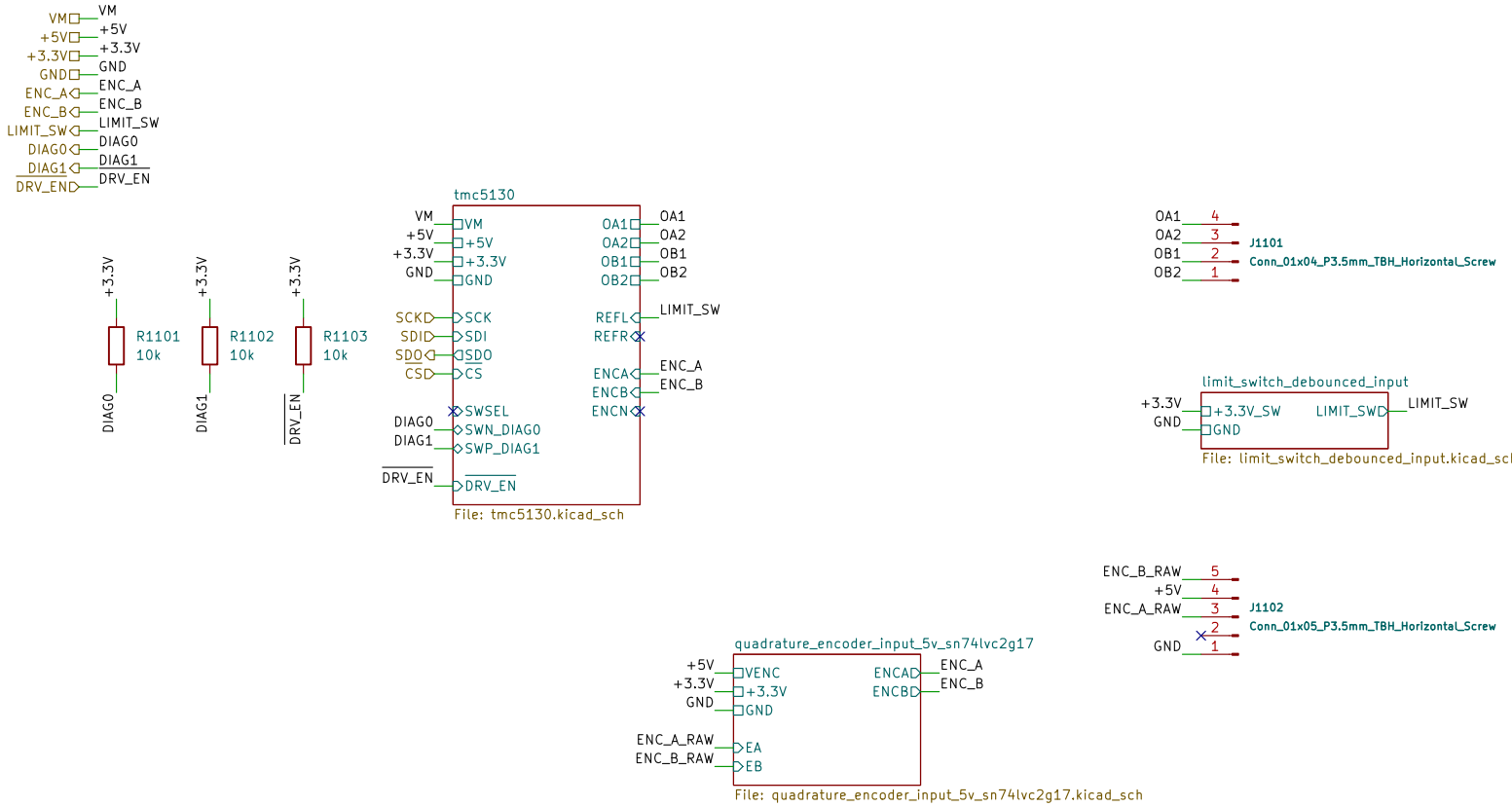
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

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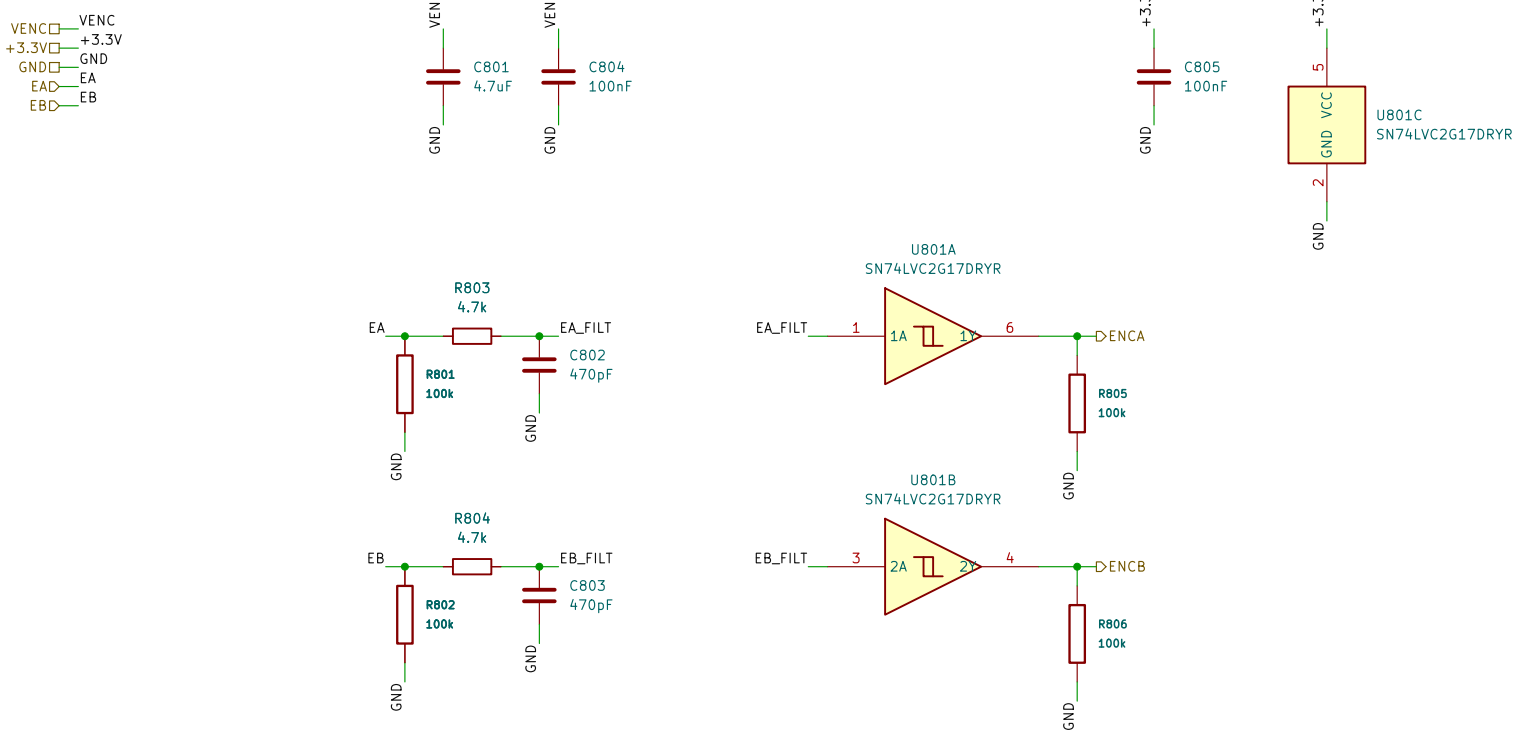
Sheet: /5v_to_3v3_tps73733_2/
File: 5v_to_3v3_tps73733_2.kicad_sch

Title: mouse-joystick-pcb

Size: A4	Date: 2026-08-19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 10/19



Janelia Research Campus		
Sheet: /motor_encoder_limit/		
File: motor_encoder_limit.kicad_sch		
Title: mouse-joystick-pcb		
Size: A4	Date: 2026-08-19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 11/19



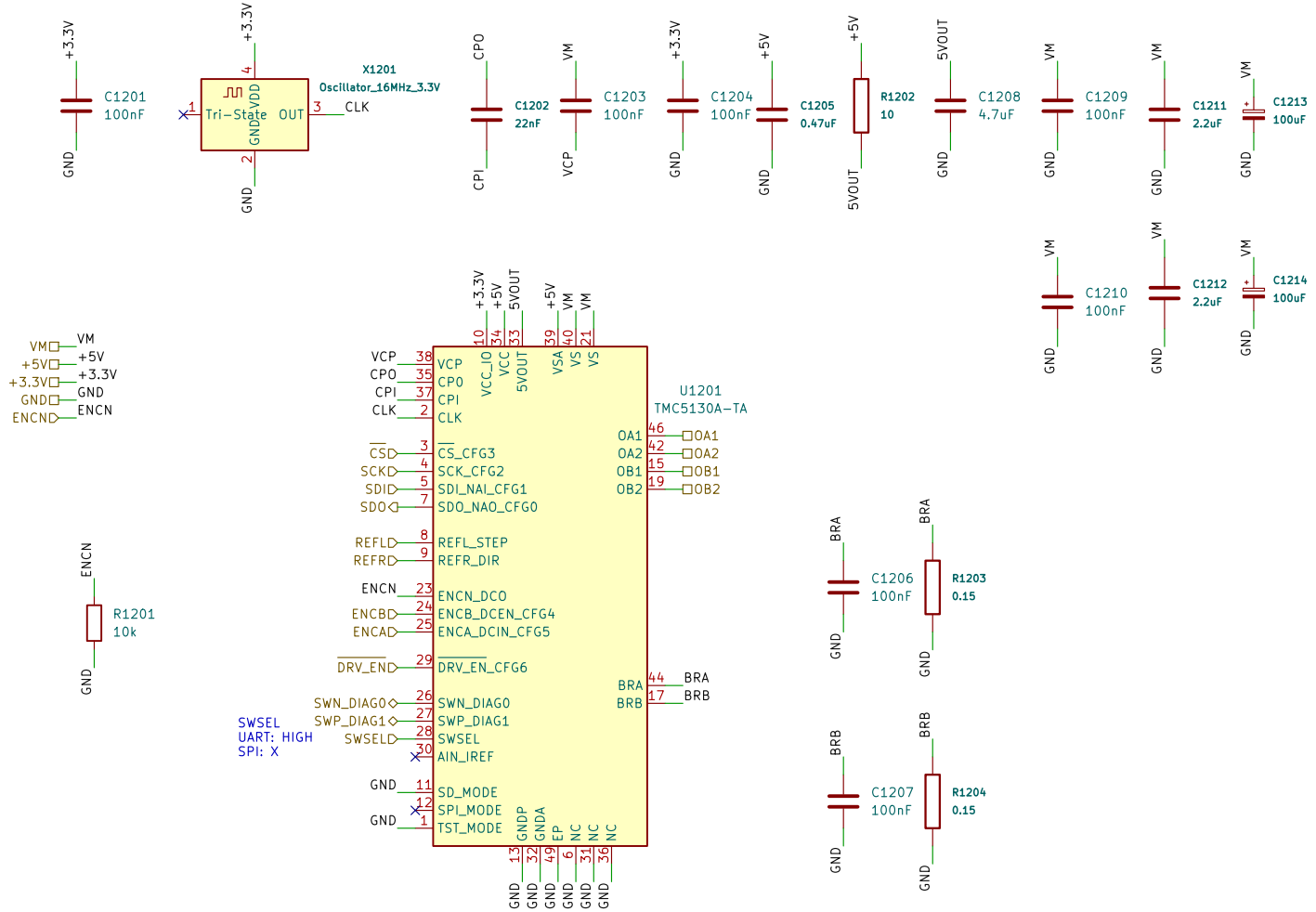
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

Janelia Research Campus

Sheet: /motor_encoder_limit/quadrature_encoder_input_5v_sn74lvc2g17/
File: quadrature_encoder_input_5v_sn74lvc2g17.kicad_sch

Title: mouse-joystick-pcb

Size: A4	Date: 2026-08-19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 8/19



TMC5130 Rsense options in stock, approx full-scale current at CS=31:
360m 1206 250mW: vsense=0 0.60A RMS / 0.86A peak; vsense=1 0.33A RMS / 0.47A peak OW-PASS-RL1206FR-070R36L
150m 1206 1W: vsense=0 1.35A RMS / 1.91A peak; vsense=1 0.75A RMS / 1.06A peak OW-PASS-RLH12FTCR150
120m 1206 750mW: vsense=0 1.64A RMS / 2.32A peak; vsense=1 0.91A RMS / 1.29A peak OW-PASS-KRL1632E-M-R120-F-T5
Set lower current in firmware; choose Rsense so normal IRUN is ~16-31.
TMC5130 TQFP thermal behavior needs validation above ~1.2A RMS continuous.

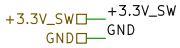
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

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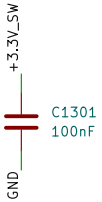
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File: tmc5130.kicad_sch

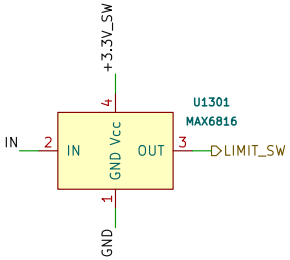
Title: mouse-joystick-pcb

Size: A4	Date: 2026-08-19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 12/19









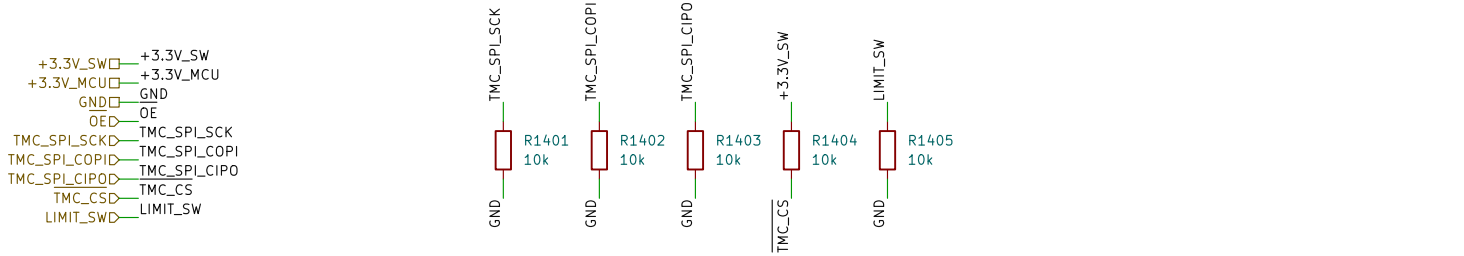
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

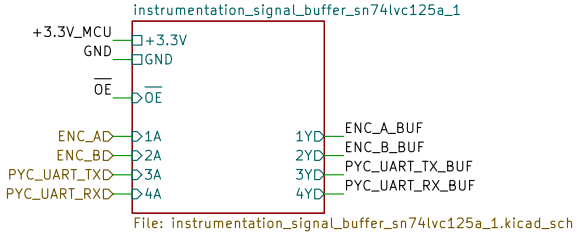
Janelia Research Campus

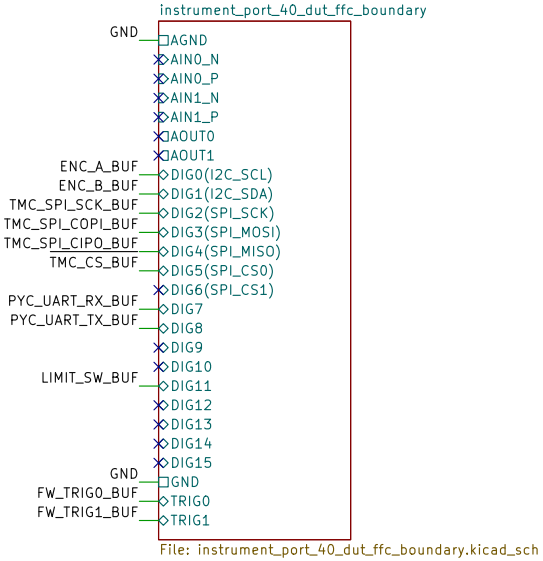
Sheet: /motor_encoder_limit/limit_switch_debounced_input/
File: limit_switch_debounced_input.kicad_sch

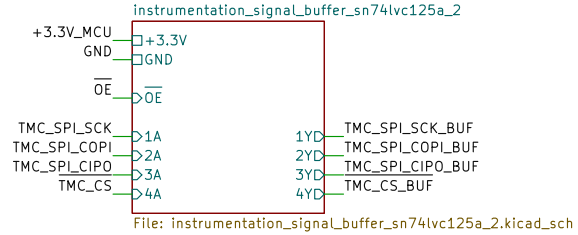
Title: mouse-joystick-pcb

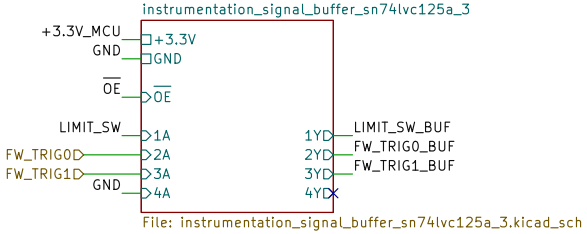
Size: A4	Date: 2026-08-19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 13/19











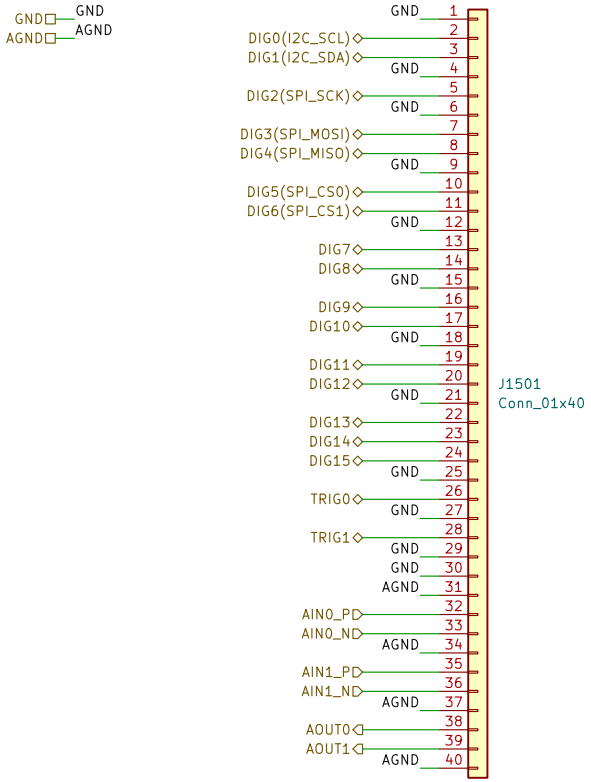
Janelia Research Campus

Sheet: /instrument_port/

File: instrument_port.kicad_sch

Title: mouse-joystick-pcb

Size: A4	Date: 2026-08-19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 14/19



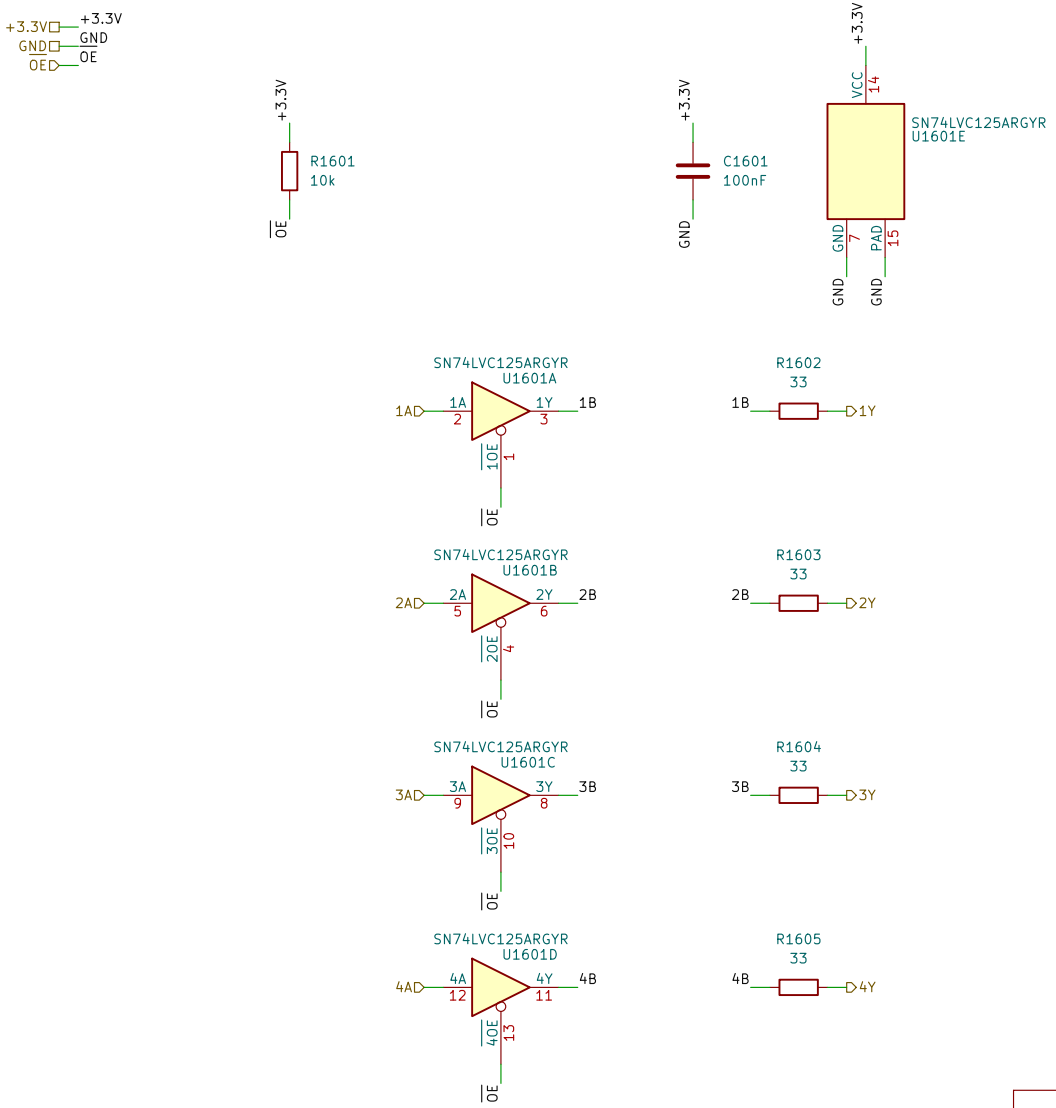
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

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Sheet: /instrument_port/instrument_port_40_dut_ffc_boundary/
File: instrument_port_40_dut_ffc_boundary.kicad_sch

Title: mouse-joystick-pcb

Size: A4	Date: 2026-08-19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 15/19



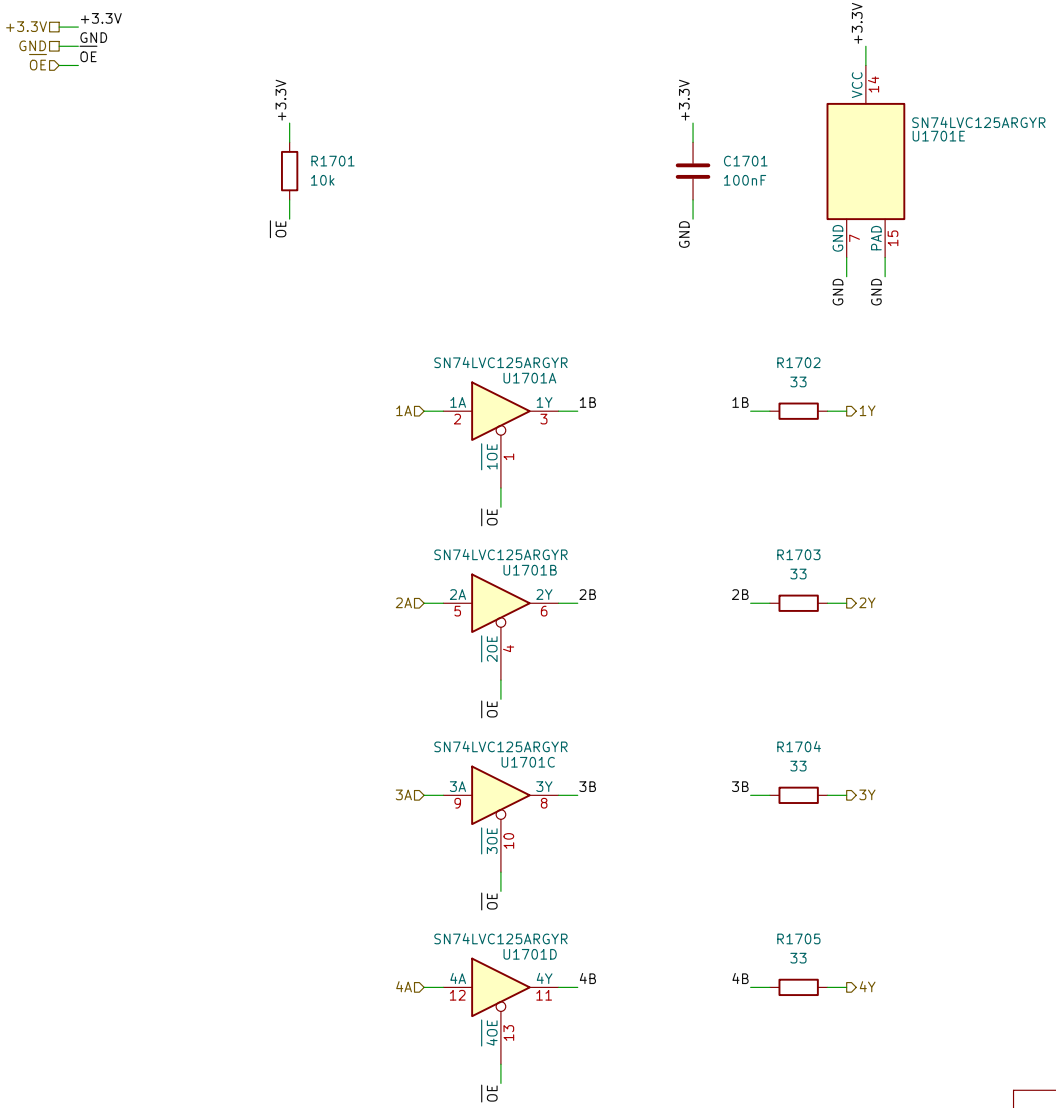
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

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Sheet: /instrument_port/instrumentation_signal_buffer_sn74lvc125a_1/
File: instrumentation_signal_buffer_sn74lvc125a_1.kicad_sch

Title: mouse-joystick-pcb

Size: A4	Date: 2026-08-19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 16/19



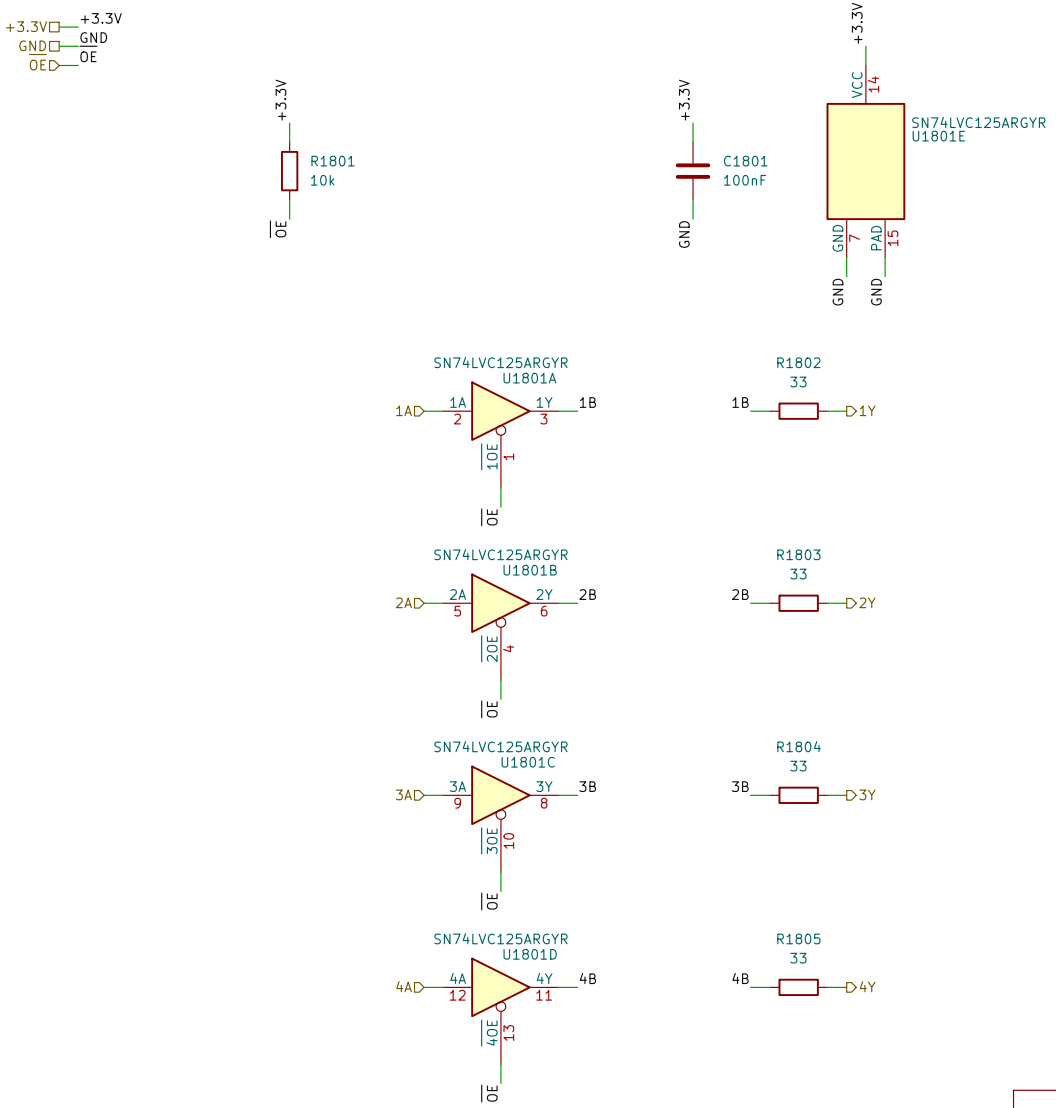
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

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Sheet: /instrument_port/instrumentation_signal_buffer_sn74lvc125a_2/
File: instrumentation_signal_buffer_sn74lvc125a_2.kicad_sch

Title: mouse-joystick-pcb

Size: A4	Date: 2026-08-19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 17/19



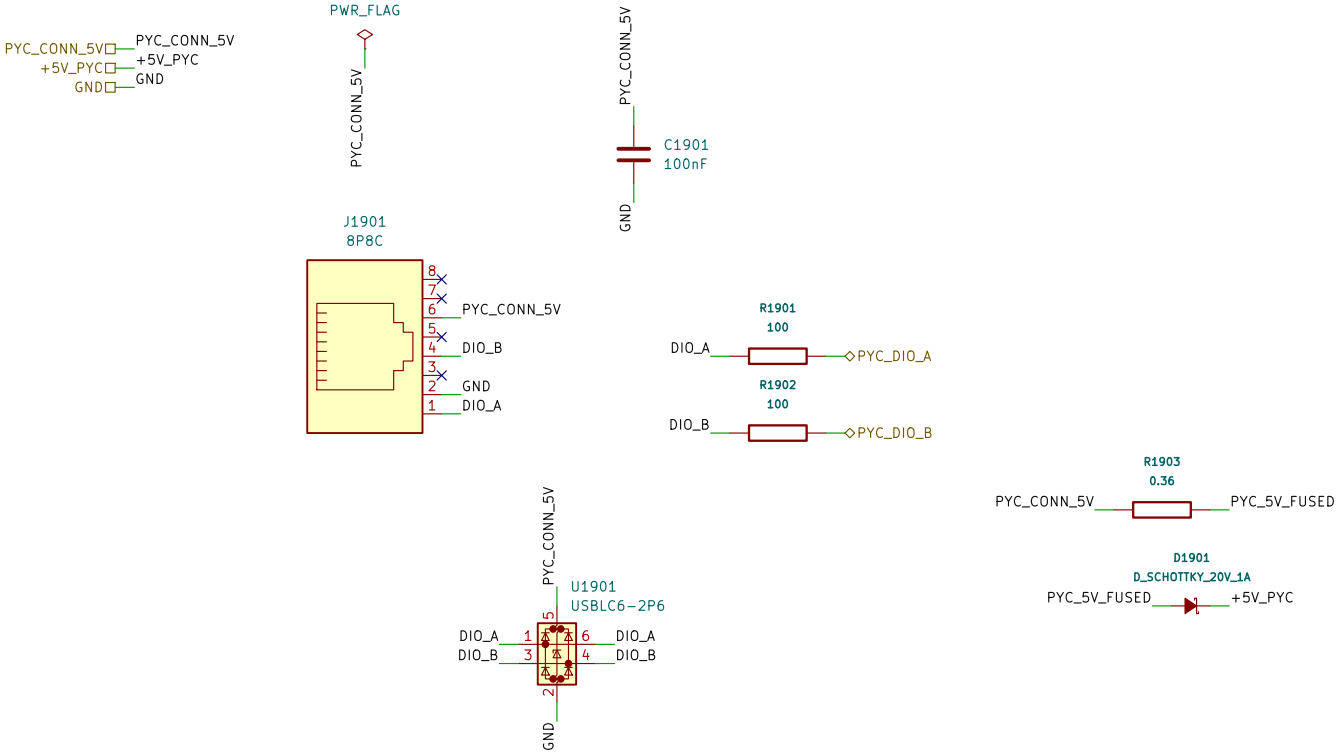
Copy or recreate old project circuits here, then save cleaned selections as design blocks.

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Sheet: /instrument_port/instrumentation_signal_buffer_sn74lvc125a_3/
File: instrumentation_signal_buffer_sn74lvc125a_3.kicad_sch

Title: mouse-joystick-pcb

Size: A4	Date: 2026-08-19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 18/19



pin 1 DIO-A = HOST TX = THIS BOARD'S RX. pin 4 DIO-B = this board's TX.
R29003 IS A POLYFUSE POSITION, NOT PROTECTION. The 0.36R part is a
jumper. Port 5 V is a shared 1.0 A that also feeds the pyControl
controller's own logic regulator, with no per-port fuse — a fault here
can reset the controller. Populate a 1206 PPTC, 250-500 mA hold.
Pins 3/5/7 are host driver outputs or a port-dependent DIO-C. Leave NC.
NOT ETHERNET. No magnetics.

Copy or recreate old project circuits here, then save cleaned selections as design blocks.

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Sheet: /pycontrol_behavior_port_rj45_boundary/
File: pycontrol_behavior_port_rj45_boundary.kicad_sch

Title: mouse-joystick-pcb

Size: A4	Date: 2026-08-19	**Rev: 0.1**
KiCad E.D.A. 10.0.5		Id: 19/19